

The Second Law of Thermodynamics Ludwig Boltzmann, who spent much of his life studying s

Let's do some combinatorics here. Imagine you have a fair coin, and you flip that coin 3 times. You define an event E

How many possible outcomes are there inside the sample space Ω (or S)?

The event E is called as the **macrostate** of the system, and each of its elements is called **microstate**. For example, the m
[width=0.7] /PhyAI/chapter2/21.png How many configurations satisfy 3 bonds containing 4 units of energy? A solid conta
[width=0.7] /PhyAI/chapter2/22.png The general formula for the multiplicity of an Einstein solid This crucial formula is t

The problem is, N and q are always very large numbers. For example, in 1 gram of Na , the number of chemical bonds is up

$$\begin{aligned}\ln N! &= \ln [N(N-1) \cdots 2 \cdot 1] = \ln N + \ln (N-1) + \cdots + \ln 2 + \ln 1 \\ &= \sum_{i=1}^N \ln i \approx \int_1^N \ln x dx \approx N \ln N - N \\ \text{[Stirling's Approximation]} \quad &\text{When } N \gg 1:\end{aligned}$$

$$\begin{aligned}\text{Recall the definition of Gamma function, for every non-negative integer } N, \text{ we always have: } \quad &\Gamma(N+1) = \int_0^{+\infty} x^N e^{-x} dx \\ &= \int_0^{+\infty} e^{N \ln x} e^{-x} dx = \int_0^{+\infty} e^{N \ln x - x} dx \\ &= \int_0^{+\infty} e^{N \ln x - x} dx \quad \text{Now we want to expand the equation by Taylor expansion, by setting an auxiliary variable } x = y + N = \\ &= N \ln N \left(\frac{y}{N} + 1 \right) - (y + N) \\ &= N \ln N + N \left(\frac{y}{N} - \frac{1}{2} \cdot \frac{y^2}{N^2} \right) - y - N \\ &= N \ln N - \frac{1}{2} \cdot \frac{y^2}{N} - N\end{aligned}$$

$$\begin{aligned}\text{After changing the variable, we obtain: } \quad &\Gamma(N+1) = \int_0^{+\infty} e^{N \ln x - x} dx = \int_{-N}^{+\infty} \exp \left(N \ln N - \frac{1}{2} \cdot \frac{y^2}{N} - N \right) dy \\ &= N^N e^{-N} \int_{-N}^{+\infty} \exp \left(-\frac{1}{2} \cdot \frac{y^2}{N} \right) dy \quad \text{Since } N \text{ approaches infinity, applying Gaussian integral result:}\end{aligned}$$

$$\begin{aligned}\text{We conclude: } \quad &\Gamma(N+1) = N^N e^{-N} \sqrt{2N} \int_{-\infty}^{+\infty} \exp \left[-\left(\frac{y}{\sqrt{2N}} \right)^2 \right] d \left(\frac{y}{\sqrt{2N}} \right) \\ &= N^N e^{-N} \sqrt{2\pi N} \quad \text{Let's apply our previous results to explore a very cool phenomenon. I have an } \mathbf{\text{interacting system}}, \text{ wh} \\ \text{[width=0.70] /PhyAI/chapter2/23.png} \quad &\text{What is the most probable configuration for energy distributing system? We have t}\end{aligned}$$

We already assumed that $N_A = 6$, $N_B = 9$, and $q = q_A + q_B = 8$, so:

The multiplicity of an interacting system is: $\Omega_{AB} = \Omega_A \cdot \Omega_B$

As you can see here, the most probable configuration of an interacting system is $q_A = 3$, $q_B = 5$, with the highest multiplici

Our calculation above seems like make sense; the "equilibrium" state of energy is the most likely reached state. Energy is w

The amount of energy stored is directly proportional to the number of chemical bonds in a solid; to put it more simply, the